Multi-Agent Validator System
Independent Evidence for Scientific Claim Verification

Abstract

False endorsement is a costly error in scientific claim checking. We test whether searching beyond the supplied citations reduces it. The validator stages four isolated roles over two evidence scopes: a sealed reader that answers a neutral question from retrieved passages before seeing the claim, a claim-visible judge over four labels (supported, contradicted, unaddressed, underdetermined), a citation auditor over the original references, and a reconciler that reports the two scopes separately instead of collapsing them into one score. Every stage fails closed: a transport or parse failure is an execution status, never a scientific verdict.

On SciFact, we compare the system with a baseline that already reads the original citations. Across 300 development claims, the baseline incorrectly endorses about 1 in 100 claims whose gold label is not support; the staged validator makes no such endorsement. Yet it finds only 43% of supported claims, compared with 57% for the baseline. A separate 300-claim held-out sample shows the same trade-off, with 38 validator requests failing closed. The reduction is *not statistically significant*. We replicate the comparison with a MiniMax family and on a frozen thirty-claim subset with four models, all closed-weight. The direction is stable: false endorsement never rises, and the cost appears as withheld contradictions. Disagreement between evidence scopes drives many abstentions. In this benchmark, every annotated gold document is already among the cited documents, limiting what independent retrieval can add. We report the negative result with a mechanism analysis and cost accounting. The system offers more conservative verdicts, but misses more supported claims.

Artifact. All tables and figures in this paper are generated from saved prediction files by the scripts in `scripts/paper/`. The corpus, splits, prompts, and frozen configurations are in the accompanying repository (Appendix F lists the commands), and the held-out split is protected by a hash freeze that refuses a run whose model, prompt, configuration, or corpus differs from the locked record.

1 Introduction

Large language models are now asked to check claims, not only to make them. In the usual pipeline a checker sees a claim and the citations offered with it, then returns a verdict. It therefore inherits the generator's evidence: if the generator cited the wrong paper, or cited a paper that supports a slightly different claim, the checker reads the same biased evidence set and cannot notice. The failure is a property of the information it is given, not a bug in the checker.

One remedy is to have the checker search for its own evidence. Recent work in retrieval-augmented factuality evaluation [Wei et al., 2024], in self-check agents [Li et al., 2026, Hirsch et al., 2026], and in retrieval-augmented verifiers [Yue et al., 2024, Zheng et al., 2024] points in that direction. Independent retrieval plus careful staging is widely assumed to help. It has rarely been tested against a strong same-evidence baseline on scientific claims, where the original citations are often correct and where an abstention can be the right answer.

This paper tests the assumption directly. We build a staged validator (**V**) that separates four roles (a sealed evidence reader, a claim-visible judge, a citation auditor, and a reconciler) over two evidence scopes: the original citations (D_0) and independently retrieved abstracts (D_1). We compare it against a same-evidence baseline (**B2**) that reads a neutral question over the original citations in an isolated reader and then compares, which is the strongest same-evidence configuration we could build without training.

We also build the full factorial around the two interventions we care about: evidence access (D_0 versus D_1) and reading policy (claim-visible versus neutral-question evidence-first).

Our preregistered primary outcome is false endorsement: the rate at which a method predicts "supported" for a claim that the gold labels do not support. The observed answer is inconclusive. On the frozen development sample the paired difference between V and B2 is -0.012 with a 95% interval of [-0.031, 0.000], which includes zero. The held-out sample repeats the pattern. The staged validator does reach zero false endorsements, but it buys that by withholding support: SUPPORT recall drops from 0.530 to 0.383 on the development sample, and on the held-out sample 38 of 300 claims fail closed entirely. Both false endorsement and support recall are needed to judge the trade-off.

Three findings frame the result.

- **Independent retrieval is not the bottleneck; scope reconciliation is.** The failure analysis shows that the dominant mode of V is not a retrieval miss but a disagreement between the D0 and D1 judgments, which the reconciler resolves conservatively into "underdetermined". When the two scopes agree, V inherits the baseline's accuracy; when they disagree, V abstains. In practice the system behaves as a disagreement detector.
- **Staged judgment buys precision, not overall accuracy.** Micro-F1 falls because recall falls faster than precision rises. Reported as F1 alone, this looks like a regression (0.747 against 0.675); reported as a precision–coverage trade-off it is a usable operating point for a conservative reader-facing tool.
- **The direction repeats across models.** We repeat the pairing on a fixed thirty-claim subset with four instruction-tuned models, all closed-weight. False endorsement never rises under staging, and the cost lands on contradiction rather than on support: REFUTE recall falls in all four models while SUPPORT recall barely moves. The subset is small, so this is a statement about direction, not significance.

No human study was run: no participant was recruited, no ethics review was obtained, and we make no claim about human trust. The four-way labels are described but not scored, because SciFact's label space is three-way and has no annotation for the distinction. The baselines are prompt-level adaptations of the named mechanisms, not reproductions of trained verifiers. Finally, our corpus is a fixed set of biomedical abstracts, so every claim of support is relative to that corpus and retrieval budget.

The contribution is a reproducible negative result with a mechanism rather than a new score: a measurement of what independent retrieval and evidence-first staging do, and do not do, to a scientific claim checker, together with the artifact that makes it checkable.

2 Related Work

2.1 Verification datasets and scientific claim checking

FEVER [Thorne et al., 2018] established the claim–evidence verification format that this paper follows, and FEVEROUS [Aly et al., 2021] extended it to unstructured and structured evidence. SciFact [Wadden et al., 2020] is the benchmark used here: it pairs scientific claims with abstracts and rationale sentences, and supplies support, refute, and insufficient-information labels. SciFact-Open [Wadden et al., 2022] extends the task to open-domain retrieval, which motivates our separation of a frozen corpus from live search; we use a frozen corpus only, and we do not claim open-domain performance. Missci [Glockner et al., 2024] targets misrepresented science and fallacy reconstruction rather than four-way claim labeling, and RAGTruth [Niu et al., 2024] targets context faithfulness of retrieval-augmented generation rather than evidence sufficiency of a checker; we cite both as boundaries of our setting rather than as benchmarks we score. For biomedical claims specifically, SciClaims [Ortega and Gomez-Perez, 2025] provides an end-to-end generative pipeline; we treat it as prior art for scientific verification with explanations and do not reproduce it. Vladika and Matthes [Vladika and Matthes, 2024] compare knowledge sources for open-domain scientific claim verification and motivate our interest in whether a different evidence source changes verdicts.

2.2 LLM verifiers, self-checking agents, and retrieval

A long line of work asks an LLM to judge claims. Chain-of-Verification [Dhuliawala et al., 2024] isolates the verification questions from the draft answer, which is the closest published analogue to our information separation and which our B2 baseline deliberately imitates. Safe [Wei et al., 2024] verifies atomic facts with search and is the reference point for our fresh retrieval arm. Self-checking agents such as MARCH [Li et al., 2026, Qwen-Applications, 2026] add information separation between proposer and checker, and blame-localization work [Hirsch et al., 2026] shows that trace-level attribution of citation errors is a developer-facing diagnostic distinct from evaluating a final conclusion for an end user; we evaluate final conclusions, and cite the trace work as the reason not to claim it. A systematic auditor for retrieval-augmented generation [Yu et al., 2026] and retrieval-augmented verifiers such as RAFTS [Yue et al., 2024] and RAV [Zheng et al., 2024] further motivate separating evidence discovery from judgment. Explainable fact checking through question answering [Yang et al., 2022] and hallucination detection methods such as SelfCheckGPT [Manakul et al., 2023] and RARR [Gao et al., 2023] address the neighbouring problem of detecting unsupported content in model output.

Little in the design is new in isolation. Atomic decomposition, blinded checking, search-based verification, and readable explanations each have substantial prior art. Our contribution is a controlled measurement of their combination against a strong same-evidence control, and a negative result about that combination.

2.3 Abstention, calibration, and selective use

A checker that cannot abstain will answer when it should not. Work on abstention [Feng et al., 2024] and on calibrated selective fact checking [Yang, 2026] formalises the trade-off we measure. Calibration theory [Guo et al., 2017] distinguishes a model's verbalised confidence from a probability fitted on held-out data; we fit a single temperature on a calibration split and report raw and calibrated numbers separately. Because a scientific "underdetermined" verdict and a system-level "defer to a reviewer" action are different things, we keep them in separate fields. On the human side, confidence and explanation are known to affect both accuracy and trust calibration in AI-assisted decisions [Zhang et al., 2020], so we do not claim that human trust calibration is unstudied; we did not run the study and we report the protocol without results.

3 Problem Formulation

3.1 Inputs and evidence scopes

Let c be an atomic scientific claim taken from a frozen report. A claim carries an exact source span, a normalised statement, and a *neutral question*: a single interrogative formulation of the claim that preserves population, comparator, units, and causal scope but does not contain the asserted answer. Two evidence scopes are available:

- D_0 — the documents the original report cited. These are the generator's evidence. A judgment over D_0 answers: *is the original citation set faithful to the claim?*
- D_1 — documents retrieved independently from a frozen corpus using the neutral question, with the claim's answer, the report, and D_0 hidden from the retrieval and reading stages. A judgment over D_1 answers: *what does independent retrieval establish?*

A verdict is always relative to a stated scope, corpus, and search budget. It is not an unrestricted declaration of scientific truth. The reconciled judgment over $D_0 \cup D_1$ answers: *what can be said once both scopes are considered?*

3.2 Four labels

Internally the system uses four labels, defined relative to a specified evidence collection:

Supported
Relevant evidence establishes the exact scoped assertion and no unresolved material counterevidence remains in the reviewed pool.

Contradicted
Relevant evidence establishes an incompatible assertion; apparent conflicts have been resolved by scope or relevance.

Unaddressed
No reviewed relevant passage answers the claim under the stated corpus and retrieval budget.

Underdetermined
Relevant evidence exists, but disagreement, imprecision, incomplete measurement, or an unsupported inference prevents resolution.

The decision order is: check execution status and claim validity; then assess relevance; if no passage addresses the claim, use *unaddressed*; if relevant evidence is materially conflicting or inconclusive, use *underdetermined*; otherwise apply support or contradiction, but only at the preserved scope. A non-significant result is not automatically proof of no effect, and a null-result claim requires its own evidence assessment.

SciFact's native label space is three-way, so the scored mapping is supported $\rightarrow$ SUPPORT, contradicted $\rightarrow$ REFUTE, and unaddressed or underdetermined $\rightarrow$ NEI. The four-way distinction is therefore *reported but not scored* against gold. We say this once here and repeat it wherever a number could be misread.

3.3 Research questions

- **RQ1 (primary, preregistered).** At a matched inference budget, does independent retrieval plus staged evidence-first judgment reduce false endorsement relative to same-evidence checking?
- **RQ2 (descriptive).** Can the system distinguish missing evidence from unresolved evidence, and how often does it defer?
- **RQ3 (deferred).** Do evidence-linked explanations improve non-experts' judgments, including when the validator is wrong? No participant was recruited; Appendix D gives the protocol and no results.
- **RQ4 (secondary).** How do accuracy, coverage, latency, and cost change across generator/checker families and across the two interventions?

4 System Design

Figure 1 shows the staged pipeline and its information boundaries. Roles are isolated functions with typed inputs; several roles may call the same underlying model in separate stateless requests. Separate roles do not imply statistically independent errors, so we measure model-family dependence rather than assume it.

Figure 1 also traces the records passed between stages. The independent reader receives the question and retrieved passages; the claim enters only after that reader has sealed its record.

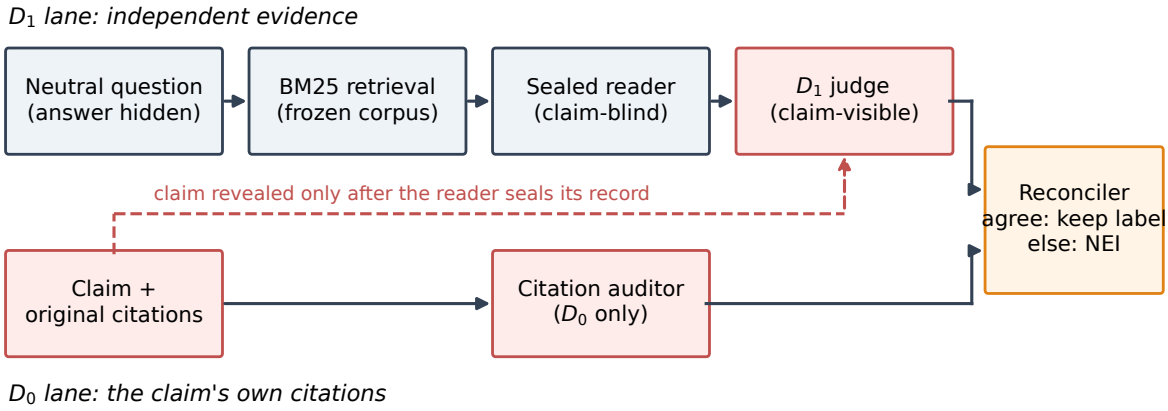


Figure 1: The staged validator. Solid arrows carry evidence; the dashed arrow is the only path by which the claim reaches the judge, and the reader's sealed record is fixed before the claim is revealed. D0 (original citations) and D1 (independent retrieval) are judged separately; the reconciler reports both scopes and never repairs one with the other. Any transport or parse failure yields a failed execution status, not a scientific label.

Table 1: The validation procedure for one claim. "Scope" is the evidence collection a step may read; no step may read outside its scope. A failure in any step produces an execution status and stops the claim.

Step	Role	Reads	Writes
1	Proposer	report	atomic claims, neutral questions
2	Gatherer	neutral question, corpus	D_1 passages
3	Reader	neutral question, D_1	sealed evidence record
4	Judge	claim, sealed record, D_1	four-way label, spans
5	Auditor	claim, D_0	D_0 label, citation adequacy
6	Reconciler	steps 4 and 5	reconciled label, conflicts
7	Renderer	reconciled record	claim card

4.1 Roles and information boundaries

Table 1 gives the procedure for one claim and the evidence scope each step may read.

Proposer.
Maps a report to atomic claims, neutral questions, asserted answers, and exact source spans. Quantities, confidence language, population, comparator, and causal scope are preserved; a conjunction is split while the relation between its parts is retained.

Independent gatherer.
Given the neutral question, a scope, and the corpus, retrieves fresh evidence. The asserted answer, the report, the original citations, and the generator's identity are hidden. Three query forms are issued per claim (open inquiry; scope and measurement; limitations or null results); the top candidates are deduplicated and reranked, and up to eight passages are kept.

Evidence reader.
Answers the neutral question from the shuffled retrieved passages and cites the passage identifiers it used. Its record is *sealed* before the claim is revealed.

Judge.
Sees the atomic claim, the sealed reader record, and the evidence passages, and assigns one of the four labels together with rationale codes, cited spans, and uncertainty reasons.

Citation auditor.
Judges the claim against D_0 alone and rates each citation's adequacy. It never sees D_1.

Reconciler.
Combines the two judgments, explains the difference, and flags invalid inference links between claims.

Renderer.
Produces plain-language claim cards from the validated structured record only. It introduces no new factual claims.

The blinding is partial: a neutral question still reveals the topic and may leak assumptions. The isolation guarantee is structural, not semantic. The reader's payload contains only the neutral question and passage text, and an assertion on the payload shape plus a test suite enforce that.

4.2 Execution and stopping

The report is generated once and frozen. Validation does not rewrite it. A claim budget of eight claims per report, two retrieval rounds, three query forms per round, and eight passages in the judge context are development defaults that were frozen before the held-out run. A second retrieval round is permitted only for unresolved questions, scope mismatch, or contradictory evidence; a run stops on exhausted budget, no new eligible documents, or a complete bounded evidence record. Every intermediate artifact is persisted, and the runner resumes at the last completed claim without mixing configurations.

The system fails closed by default. When the model returns malformed JSON, cites a passage it was not given, or the endpoint fails, the system records an execution status of `failed` or `timeout` and a placeholder label that is a schema token, never model text. These rows are excluded from scored metrics and reported separately. They are never silently remapped into a scientific verdict.

4.3 Reconciliation

Agreement between scopes copies the shared label. A disagreement, or a failed D0 audit, resolves to *underdetermined* and hence to NEI on the scored label space. We do not repair citation faithfulness with outside evidence: a claim can be unsupported by its own citation and supported elsewhere, and the two facts are reported separately.

5 Implementation

5.1 Records and reproducibility contract

The implementation emits a machine-readable prediction file and an immutable run sidecar for every condition. The run record carries a run identifier, split, seed, model identifier and revision, prompt-bundle hash, corpus hash, configuration hash, timestamps, token totals, and an execution status; the sidecar adds the prompt files and their individual hashes. Predictions carry the claim identifier, the predicted label, the four-way label where applicable, the evidence scope, the rationale codes, and the execution status. Schema validation is enforced before a file is accepted.

The corpus is frozen by a SHA-256 hash over the canonicalised abstract corpus, and the split manifests are locked separately. The held-out split is protected by a freeze record containing the model identifier, the corpus hash, and per-method configuration and prompt hashes; a held-out run refuses to start unless the observed hashes match the frozen ones. This is what makes "we ran the held-out split once" a checkable statement rather than an assurance.

5.2 Retrieval

Retrieval is BM25 (Okapi, $k_1 = 1.5$, $b = 0.75$, epsilon idf floor) implemented in pure Python over the frozen abstract corpus, with a fixed alphanumeric lower-case tokenizer and no stemming. For each neutral question three query templates are used; the top 20 candidates per query are merged, deduplicated, and reranked by maximum BM25 score, and at most eight passages are passed to the judge. Adjacent sentences of one document are kept together where needed to resolve negation or units. Dense retrieval and reciprocal-rank fusion were prepared but are not enabled in the product path; they are development-time comparisons, and we do not report a dense retrieval result we did not measure.

5.3 Models and harness

We evaluate two model families on the full samples to satisfy RQ4 and to protect the negative result from being an artefact of one model's behaviour, and we repeat the pairing on a wider set of models over the fixed subset of Section 7.4.

GPT-6 LUNA.
A hosted instruction-tuned model accessed through an OpenAI-compatible chat endpoint. It is the model behind the powered comparison, the held-out run, and the cost accounting.

MINIMAX M3.1 FLASH.
A second, architecturally unrelated instruction-tuned model, accessed through the local agent runtime. To keep the repository's endpoint policy intact we expose that runtime through a small loopback bridge that implements the OpenAI chat completions interface, one fresh session per request. A fresh session per request matters: a shared session would carry earlier claims into later ones and break the information-isolation guarantee the validator is measured on. The harness adds a fixed agent preamble and tool definitions to every request; the preamble is byte-identical across conditions, so it cancels in every comparison, but it inflates the prompt-token counts. That is why cost is reported for the Luna family only, and why the MiniMax family contributes no cost row.

WIDER SET.
The sweep adds three more models on that subset: GPT-6 Astra, GPT-6 Sol and GPT-5.6 Terra. They use the same hosted API as the Luna family, so their cost and token accounting match the Luna family's.

All models run with temperature 0 where the endpoint accepts it, and with the same prompts, frozen corpus, claim sample, and budgets. The prompts are versioned text files under `prompts/`; their hashes enter the freeze record.

5.4 Failure handling and the retry policy

Three different things can go wrong, and the paper keeps them apart.

1. **Transport failure.** The endpoint refuses the connection or returns a retryable status. The client retries with exponential backoff.
2. **Protocol failure.** The model returns text that is not the requested JSON, cites a passage it was not given, or returns a label outside the allowed set. The parser rejects it. This is a *parse* failure, not a scientific disagreement.
3. **Scope conflict.** Both stages completed and disagree. This is a scientific outcome: the reconciled verdict is *underdetermined*, which maps to NEI.

The first two produce an execution status of `failed` and are excluded from the scored metrics; the third is scored normally. Because a failed request is not evidence about the claim, the runner retries it, up to four attempts per claim, and drops the failed row from the prediction file between attempts so that

Table 2: The condition matrix. "Adaptation" means a prompt-level implementation of the named mechanism, never a reproduction of a trained system. The 2×2 factorial is B1/B2 (evidence D_0) against B3/V-D1 (evidence D_1), crossed with claim-visible against neutral-question evidence-first reading.

ID	Condition	Evidence and reading policy
B0	Prior-knowledge judge	No evidence; claim only
B1	Atomic claim, claim-visible judge	D_0; claim-visible
B2	Neutral question, isolated reader, compare	D_0; evidence-first
B3	Fresh retrieval, direct judge	D_1; claim-visible
V-D1	Fresh retrieval, isolated reader, judge	D_1; evidence-first
V	Full staged validator	$D_0 \cup D_1$; reconciled
O	Evidence oracle (diagnostic)	Gold rationale sentences

the resume re-claims exactly those claims. The policy is applied identically to every condition and every model family on the development sample.

Because dropping failed rows could flatter a method that fails on the hard claims, we report a second, stricter accounting in which every submitted row is scored with the label the method recorded for it. On the fail-closed paths that label is the conservative NEI, so the strict accounting can only hurt a method. Table 5 shows that the primary comparison does not depend on the choice.

One asymmetry remains: the held-out run was always planned as a single execution and does not use the retry policy, so its fail-closed count includes unrecovered execution failures.

5.5 Prompt contracts

Each role has a fixed output contract, expressed in the prompt and enforced by the parser. The reader must return an answer and the identifiers of the passages it used; the judge must return one of four labels, rationale codes, cited span identifiers, and uncertainty reasons; the citation auditor must return a label plus a per-citation adequacy rating from {adequate, inadequate, unverifiable} and a defect list. The parsers reject a label outside the allowed set, a cited identifier that was not supplied, and any missing field. Appendix A states each contract; the prompt text itself is in the repository, versioned and hashed.

6 Experimental Setup

6.1 Data and splits

We use SciFact [Wadden et al., 2020]. The corpus is the released set of biomedical abstracts, frozen by hash. The released training claims are split into a calibration subset of 100 claims, reserved before any prompt work, and a development pool of 709 claims. The public development split (300 claims) is used as a *local* held-out evaluation set; it is not an untouched community test set, and we name it accordingly throughout. Official test labels are not used.

Two samples are frozen by manifest: a 300-claim development sample drawn with a fixed seed from the training pool (`dev300_seed42.json` under `data/manifests/`), used for all development and prompt decisions, and a 300-claim evaluation sample consisting of SciFact's public development split, held out locally and used once under the freeze record. Both samples group claims by cited-document family so that paraphrases and derivative claims do not straddle the comparison.

6.2 Conditions

All conditions share the frozen reports, the corpus snapshot, the claim sample, the scoring code, and the prompt files. They differ only in the intervention named in Table 2.

B0 measures how much the model answers from parametric knowledge alone and acts as a floor. O supplies the gold rationale sentences to the same claim-visible judge and is a diagnostic ceiling, not a deployable competitor. V-D1 is the D1 arm of V, recovered from the V predictions by a mapping step rather than a second model run, so it inherits V's failure modes by construction.

6.3 Metrics

Verification quality is measured by per-class precision, recall and F1, plus micro and macro F1 on the native three-way label space. The primary metric is *false endorsement*: the number of claims predicted SUPPORT whose gold label is not support, divided by the number of gold non-support claims. We report the second denominator (false endorsements over all predicted SUPPORT claims) as well, because the two answer different questions. Coverage is the fraction of submitted claims that produced a scored prediction; fail-closed rows are counted and reported, never dropped silently. Calibration is measured with multiclass Brier score, negative log-likelihood, and expected calibration error, fitted with a single temperature on the calibration split only. Cost is reported as calls per claim, measured tokens per call, and per-call latency, because a resumed run keeps only its last invocation's usage record and per-claim token totals are therefore not recoverable.

6.4 Statistics and preregistration

One primary comparison was preregistered before the held-out run: V versus B2 on false endorsement with the gold non-support denominator. The interval is a paired cluster bootstrap over cited-document families, 2000 resamples, 95% percentile interval; a claim enters the comparison only when every compared method produced an execution-ok, gold-labelled row, and dropped claims are counted. Secondary comparisons within a declared family are Holm-adjusted; everything else is labelled exploratory. The factorial comparison is used to interpret the mechanism, not to claim additional significance.

7 Results

7.1 Primary comparison

Table 3 gives the development results and Table 4 the preregistered paired comparison. In the Luna family, B2 reaches micro-F1 0.747 with false endorsement 0.011, while V reaches micro-F1 0.675 with false endorsement 0.000 on all 268 scored claims. The paired difference in false endorsement is -0.012 with 95% interval [-0.031, 0.000] over 268 paired claims and 226 families: the interval includes zero, so the primary comparison is null. The MiniMax family behaves the same way: B2 reaches micro-F1 0.870 with false endorsement 0.033, V reaches 0.830 with 0.016, and the paired difference is -0.016 ([-0.046, 0.011]) over 300 paired claims, again including zero.

Figure 2 shows the false-endorsement rate for each completed development condition. Missing bars are unrun conditions, as listed in Table 3.

Fail-closed rows are excluded from the scored metrics above, so a method could in principle look better by failing on the hard claims. Table 5 therefore repeats the two primary conditions under both accountings. Under the all-rows accounting every submitted claim is scored with the label the method recorded for it, which is the conservative NEI on every fail-closed path in this repository. The comparison does not depend on the choice: V's false endorsement is zero under both accountings on the development sample, where the retry policy recovered every failed request. On the held-out sample, which ran once without that policy, the all-rows accounting lowers V's micro-F1 from 0.618 to 0.583 while its false endorsement stays at zero.

The held-out sample reproduces the pattern. B2 reaches micro-F1 0.760 with false endorsement 0.017, and V reaches micro-F1 0.618 with false endorsement 0.000 on 262 scored claims with 38 failures. The paired difference is -0.013 ([-0.033, 0.000]), again including zero.

Table 3: Development sample ($n = 300$ claims, seeded draw). Method codes: B0 prior knowledge, B1 D0 claim-visible, B2 D0 evidence-first, B3 D1 claim-visible, V-D1 D1 evidence-first, V reconciled over both scopes, O gold rationales. FE is false endorsement on the gold non-support denominator; the complete definition and the second denominator are in Appendix B. Fail-closed rows are excluded from the scored metrics and accounted for in Table 5.

Method	n	micro-F1	macro-F1	FE	SUPPORT R
GPT-6 Luna					
B0	300	0.557	0.517	0.448	0.872
B1	300	0.840	0.838	0.049	0.838
B2	300	0.747	0.755	0.011	0.530
B3	300	0.790	0.787	0.055	0.795
V-D1	268	0.743	0.745	0.019	0.598
V	268	0.675	0.673	0.000	0.383
O	300	0.857	0.860	0.016	0.735
MiniMax M3.1 Flash					
B1	300	0.863	0.856	0.093	0.966
B2	300	0.870	0.867	0.033	0.855
V-D1	300	0.853	0.851	0.033	0.829
V	300	0.830	0.837	0.016	0.692

Table 4: Paired comparison of V against B2 on false endorsement, by family and split. A claim enters once both methods produced an execution-ok row; p is the exact McNemar test on the discordant pairs, and the interval is a paired cluster bootstrap over cited-document families.

Model	Run	n	non-supp.	B2 only	V only	McNemar p	ΔFE [95% CI]
GPT-6 Luna	development	268	161	2	0	0.50	-0.012 [-0.031, 0.000]
GPT-6 Luna	development, second run	300	183	2	0	0.50	-0.011 [-0.029, 0.000]
GPT-6 Luna	held-out	262	156	2	0	0.50	-0.013 [-0.033, 0.000]
MiniMax M3.1 Flash	development	300	183	5	2	0.45	-0.016 [-0.046, 0.011]

The null has structure. V achieves near-perfect precision on the endorsement class (SUPPORT precision 1.000 in the Luna family, 0.964 in the second) and pays for it in recall. The system is a conservative checker, not a better one. Micro-F1 alone reads as a regression, and false endorsement alone reads as a success; the paired interval is what distinguishes the two readings.

7.2 The 2×2 factorial

Table 6 separates the two interventions. **Evidence access** is the row difference (B1 versus B3, and B2 versus V-D1); **reading policy** is the column difference (B1 versus B2, and B3 versus V-D1). The table shows three effects.

First, *claim-visible judging beats evidence-first judging* on the scored label space: B1 (D0, claim-visible) beats B2 (D0, evidence-first) on micro-F1 (0.840 versus 0.747), and B3 (D1, claim-visible) beats V-D1 (D1, evidence-first) (0.790 versus 0.743). This is the opposite of the intuition behind evidence-first staging. Reading the evidence without the claim in view loses information that the judge needs; the sealing buys isolation, not accuracy.

Second, *evidence access barely moves the evidence-first cell*: V-D1's false endorsement (0.019) is close to B2's (0.011) even though V-D1 has access to a larger evidence pool, and in the MiniMax family the two are equal (0.033 against 0.033). Independent retrieval introduces documents that the original claim was never calibrated against, and it does not buy accuracy.

Third, *the full V is more conservative than either arm*, because the reconciler resolves scope disagreement into abstention rather than choosing a side. V's zero false endorsement is thus a property of the reconciliation policy, not of the retrieval.

The factorial is a Luna-family result. The MiniMax family contributes only B1, B2, V-D1 and V,

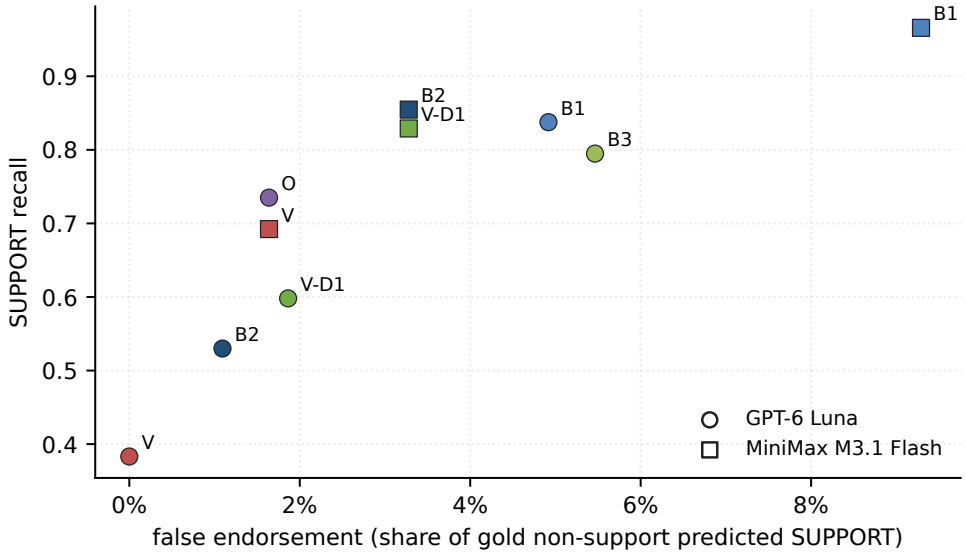


Figure 2: False endorsement on gold non-support claims by condition and model. Lower is better, but the rate must be read alongside support recall and the paired intervals. Missing conditions were not completed.

and those rows do not repeat the claim-visible advantage: there B1 reaches micro-F1 0.863 against B2's 0.870, and B1 is the less conservative arm (false endorsement 0.093 against 0.033). We therefore state the factorial as a finding about this benchmark under one model family, and treat its replication as open rather than asserting it.

Table 7 breaks the two primary conditions down by class. The pattern is the same in both families. The staged validator raises the precision of the endorsement class (from 0.969 to 1.000 in the Luna family, 0.943 to 0.964 in the second) by pushing almost everything it is unsure about into NEI, and it pays for that with recall (0.530 against 0.383, and 0.855 against 0.692) and with lower NEI precision (0.628 against 0.560, and 0.847 against 0.722), because it abstains on far more claims and many of those claims are supported.

7.3 Model-family transfer

Table 3 reports both families. The pattern (V more conservative than B2, lower SUPPORT recall, false endorsement at or near zero, primary interval including zero) is reproduced under a second model family reached through a different harness: the paired false-endorsement difference is -0.012 ([-0.031, 0.000]) for the Luna family and -0.016 ([-0.046, 0.011]) for the second. The two families differ in absolute quality: the MiniMax family is a markedly better verifier on this benchmark (B2 micro-F1 0.870 against 0.747), and the direction of the V effect is the same in both. This is consistent with the staged design producing a conservative operating point, although two model families cannot establish that as a general effect. We do not read it as evidence about the absolute quality of either model; the two families are not matched for size, training data, or capability, and the comparison is between families of systems, not between models of equal strength. We also cannot separate the MiniMax family's model prior from its design: its prior-knowledge condition (B0) and its claim-visible D1 condition (B3) did not complete before the free endpoint that served it was withdrawn, so the MiniMax family contributes its B1, B2 and V rows rather than the full factorial. It corroborates the preregistered comparison and not the wider factorial.

7.4 The shift across four models

Two families answer the preregistered question, but not whether the conservative shift belongs to the design or to the two models we happened to run. To check the direction more widely we repeated the

Table 5: Sensitivity to the fail-closed accounting. "Scored only" excludes execution failures from the metrics; "all rows" scores every submitted claim with the label the method recorded.

Method	accounting	n	micro-F1	FE	SUPPORT R
GPT-6 Luna					
B2 (dev300)	scored only	300	0.747	0.011	0.530
B2 (dev300)	all rows	300	0.747	0.011	0.530
V (dev300)	scored only	268	0.675	0.000	0.383
V (dev300)	all rows	300	0.640	0.000	0.350
B2 (dev300rerun)	scored only	300	0.747	0.011	0.573
B2 (dev300rerun)	all rows	300	0.747	0.011	0.573
V (dev300rerun)	scored only	300	0.687	0.000	0.427
V (dev300rerun)	all rows	300	0.687	0.000	0.427
B2 (heldout300)	scored only	300	0.760	0.017	0.581
B2 (heldout300)	all rows	300	0.760	0.017	0.581
V (heldout300)	scored only	262	0.618	0.000	0.368
V (heldout300)	all rows	300	0.583	0.000	0.315
MiniMax M3.1 Flash					
B2 (dev300)	scored only	300	0.870	0.033	0.855
B2 (dev300)	all rows	300	0.870	0.033	0.855
V (dev300)	scored only	300	0.830	0.016	0.692
V (dev300)	all rows	300	0.830	0.016	0.692

same B2-versus-V pairing on a frozen subset of the development sample: ten SUPPORT, ten REFUTE, and ten NEI claims, selected with seed 42 before any model was run. In total four instruction-tuned models took part, all closed-weight. All used the same prompts, corpus, retrieval budget, and scoring code. Table 8 lists them.

The subset is small and we read it that way. Thirty claims can resolve a difference of roughly ten percentage points, not the one-point false-endorsement rate the full sample was sized for. Most of the paired intervals in Table 8 therefore collapse to a point at zero, and we do not test these rows for significance. What the subset can show is where on the label set the conservative shift lands.

Two regularities hold across the rows. False endorsement never rises: ΔFE is zero in four of the four models, negative in none, and positive in none; none of the four intervals exclude zero. And the abstention lands on contradiction rather than on support: REFUTE recall is lower under V in four of the four models and unchanged in none, while SUPPORT recall is unchanged in three models and lower in one. Micro-F1 falls in four of the four.

The closed-weight rows make the pattern visible. Under B2 each of them identifies most refuted claims; under V each of them identifies half. Support recall moves by at most one claim in the same rows. On the full sample the loss is not exclusive to REFUTE (Table 7 shows both classes falling), so the subset is better read as showing where the shift is concentrated than as a separate finding. The mechanism is the same one Section 7.9 describes: when the two scopes disagree, the reconciler prefers *underdetermined*, and disagreement is more common exactly where a citation states the opposite of the claim.

Figure 3 draws the same shift per model.

The claim-level records say whether the models lose the same claims or different ones. Of the nine refuted claims that the baseline identifies, four are withheld under V by every model that called them refuted, five by none, and none by some but not all: the recall loss is a handful of specific claims, not a diffuse erosion. Support loss is concentrated the same way: of the seven supported claims the baseline finds, six are kept by every model and one is lost by some but not all. No model gains a refutation the baseline missed, and the reverse gain for support occurs in one model–claim pair. The models also agree on the labels themselves: all four give the same B2 label on 24 of the thirty claims and the same V label on 27.

The sweep also bounds how much of the result can be a model artefact. Models differ in absolute quality: B2 micro-F1 ranges from 0.800 to 0.867 across the four, and that ordering does not predict which

Table 6: Factorial ablation on the development sample. FE is false endorsement on the gold non-support denominator. V-D1 is derived from the V predictions, not re-run.

Cell	Evidence	Reading policy	micro-F1	FE
GPT-6 Luna				
B0	none	prior knowledge	0.557	0.448
B1	D0	claim-visible	0.840	0.049
B2	D0	neutral / evidence-first	0.747	0.011
B3	D1	claim-visible	0.790	0.055
V-D1	D1	neutral / evidence-first	0.743	0.019
V	D0 ∪ D1	reconciled	0.675	0.000
O	gold rationales	claim-visible	0.857	0.016
MiniMax M3.1 Flash				
B1	D0	claim-visible	0.863	0.093
B2	D0	neutral / evidence-first	0.870	0.033
V-D1	D1	neutral / evidence-first	0.853	0.033
V	D0 ∪ D1	reconciled	0.830	0.016

Table 7: Per-class precision, recall, and F1 on the development sample, with the confusion counts.

Method	class	P	R	F1	TP	FP	FN
GPT-6 Luna							
B2	SUPPORT	0.969	0.530	0.685	62	2	55
B2	REFUTE	0.875	0.790	0.831	49	7	13
B2	NEI	0.628	0.934	0.751	113	67	8
V	SUPPORT	1.000	0.383	0.554	41	0	66
V	REFUTE	0.941	0.627	0.753	32	2	19
V	NEI	0.560	0.982	0.713	108	85	2
MiniMax M3.1 Flash							
B2	SUPPORT	0.943	0.855	0.897	100	6	17
B2	REFUTE	0.800	0.903	0.848	56	14	6
B2	NEI	0.847	0.868	0.857	105	19	16
V	SUPPORT	0.964	0.692	0.806	81	3	36
V	REFUTE	0.944	0.823	0.879	51	3	11
V	NEI	0.722	0.967	0.827	117	45	4

models lose the most under staging. What does not vary is the direction of the change. A design whose only effect were to make a weak model abstain more would not produce that regularity.

7.5 Where the abstentions come from

The RQ2 descriptive split is available because V stores its four-way label alongside the mapped native label. On the development sample V predicts NEI on 225 of 300 claims (0.75 of the sample), and those NEI predictions split into *unaddressed* (no retrieved passage addresses the claim, 60 claims) and *underdetermined* (relevant evidence exists but does not resolve, 165 claims). Figure 4 shows the distribution.

Whether the two scopes agreed is more informative than the four-way label. V records a `d0_d1_conflict` code when the D_0 and D_1 four-way verdicts differ, and the reconciler then abstains by construction. Table 9 conditions accuracy on that split. When the scopes agree, V's accuracy matches the baseline's (0.828 against 0.828 on the development sample); when they differ, V abstains on every claim and its accuracy falls to the share of NEI in that subset (0.309). The shape repeats on the held-out sample (0.842 against 0.871 when the scopes agree, 0.100 when they differ) and in the MiniMax family (0.925 against 0.911).

Of the 225 NEI verdicts, 55 are scope conflicts. The other 88 are agreed negatives: 60 unaddressed

Table 8: B2 versus V on the frozen thirty-claim subset. Cells after the interval are "B2 / V". ΔFE is the paired difference in false endorsement with its 95% interval; a positive value means V endorses more. The subset is descriptive, not powered: read it for direction, not for significance.

Model	Weights	ΔFE [95% CI]	SUPPORT R	REFUTE R	micro-F1
GPT-6 Astra	closed	+0.000 [0.000, 0.000]	0.700 / 0.600	0.900 / 0.500	0.833 / 0.700
GPT-6 Sol	closed	+0.000 [0.000, 0.000]	0.700 / 0.700	0.800 / 0.500	0.833 / 0.733
GPT-5.6 Terra	closed	+0.000 [0.000, 0.000]	0.700 / 0.700	0.900 / 0.500	0.867 / 0.733
GPT-6 Luna	closed	+0.000 [0.000, 0.000]	0.500 / 0.500	0.900 / 0.500	0.800 / 0.667

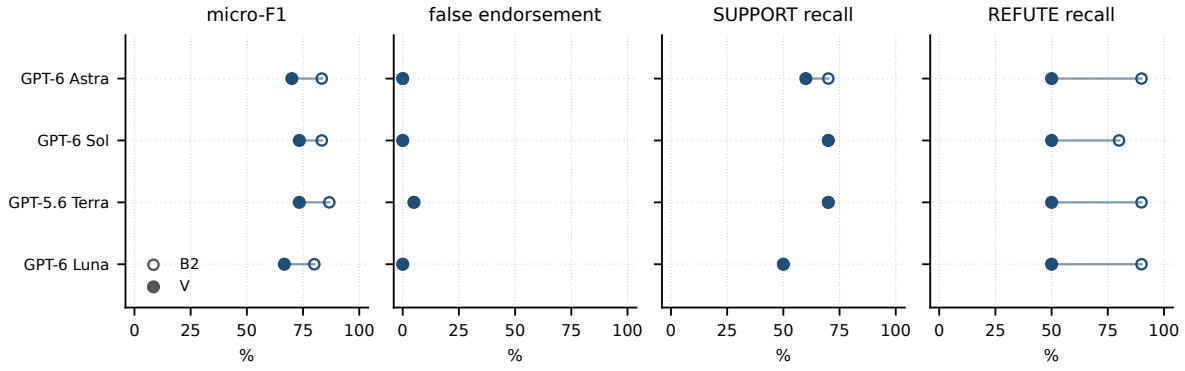


Figure 3: B2 to V on the thirty-claim subset, per model. Hollow marks are B2, filled marks are V. The conservative shift costs REFUTE recall in all four models while SUPPORT recall and false endorsement barely move; the largest false-endorsement rate in any row is 5%.

and 28 underdetermined.

Retrieval is not returning nothing. The validator abstains because two readers of the same corpus disagreed, and, less often, because both readers found the evidence inconclusive. In both cases the pipeline acts as a disagreement detector and reports its own uncertainty; the price is coverage.

7.6 Evidence acquisition

Table 10 reports what independent retrieval actually bought. Recall@k measures how often the gold rationale documents appear in the retrieved set; overlap measures how much of D_1 was already in D_0; unique yield measures how often retrieval found a relevant document the original citations missed.

These numbers are unflattering to the hypothesis that independent retrieval should help. Retrieval finds the annotated documents most of the time (recall@k = 0.85 over the 179 claims that have annotated evidence), but every annotated document it finds was also cited by the report: the unique yield is 0.00. The limit is in the benchmark, not in the retriever. In SciFact the cited document set contains the annotated document set, so a retrieval system that searches the same corpus cannot find *new gold evidence*; it can only add unannotated context, and 93% of what it returns is unjudged. The independent arm therefore cannot repair a wrong citation on this corpus, because a wrong citation is, by construction, a document without a rationale. More retrieval cannot resolve the D0/D1 conflict, and this property bounds every claim in this paper.

7.7 Calibration, selective use, and cost

Because the judge emits labels but no probabilities, we report two things. First, calibration of the one condition that verbalises confidence (B1): a single temperature fitted on the calibration split, giving Brier score, NLL, and ECE on the development split (Table 11 and Appendix C); Figure 5 plots the reliability curve for that condition. Second, a selective-use analysis that does not require probabilities: accepting a claim only when the two evidence scopes agree, and measuring error among accepted claims as a function

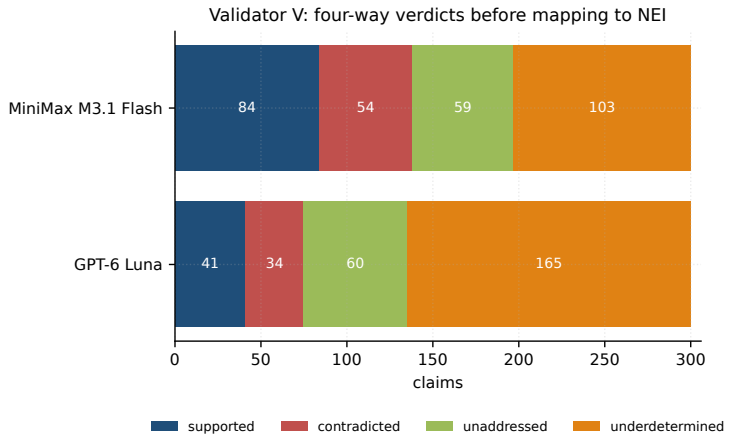


Figure 4: V's four-way verdicts on the development sample, before the mapping onto SciFact's three-way label space. The split between *unaddressed* and *underdetermined* is descriptive: SciFact's gold cannot score it.

Table 9: Accuracy conditioned on whether the two evidence scopes agree. "Differ" means the run recorded `d0_d1_conflict`; V abstains on every such claim by construction, so the column measures how often NEI happens to be the gold label. Fail-closed rows are excluded.

Scopes	n	B2 acc.	V acc.	V NEI
GPT-6 Luna, development				
same four-way label	163	0.828	0.828	0.540
both no verdict, different kind	50	0.600	0.580	1.000
different scored label	55	0.618	0.309	1.000
GPT-6 Luna, held-out				
same four-way label	139	0.871	0.842	0.496
both no verdict, different kind	73	0.603	0.548	1.000
different scored label	50	0.700	0.100	1.000
MiniMax M3.1 Flash, development				
same four-way label	214	0.911	0.925	0.355
both no verdict, different kind	36	0.861	0.917	1.000
different scored label	50	0.700	0.360	1.000

of coverage. Figure 6 shows the result for both families. The disagreement-based acceptance rule reaches low error at moderate coverage, which is the usable form of the system's conservatism.

Table 12 reports model calls, measured tokens per call, and latency for the Luna family. V issues three calls per claim against B2's two, because it runs a sealed reader, a claim-visible judge, and a citation auditor. A B2 call measures about 605 tokens and a V call about 2,579, so V's three calls per claim cost roughly six times B2's two (Figure 7 plots quality against per-call prompt size). The MiniMax family's cost is not in the table: its harness prepends a fixed agent preamble to every request, so a per-call figure would measure the harness more than the task, and the endpoint that served the model was withdrawn before the probe could be run. The call ratio does carry over, because the three stages are the same three stages. In exchange for that cost V produces two scoped judgments instead of one. A reader-facing deployment that only needs a single verdict is better served by B2; the value of V is that its two scopes are separately auditable.

7.8 Failure analysis

We classify the errors of the scored rows against a taxonomy fixed before the analysis: false endorsement, missed support, missed contradiction, and other label errors, plus a separate count of execution failures that are never scored as scientific verdicts. The classification uses only fields the runner records, so it is

Table 10: Evidence acquisition for the independent retrieval arm on the development sample. Recall@k is over claims that have at least one gold rationale document; overlap is the mean fraction of retrieved documents that were also cited by the report.

Quantity	Value
Claims scored	300
Claims with annotated evidence	179
Passages retrieved per claim (k)	8
Recall@k (annotated documents found)	0.849
Precision@k (retrieved documents annotated)	0.074
Overlap with D_0 (retrieved documents also cited)	0.096
Unique yield (annotated found but not cited)	0.000
Refuting yield (contradicting documents found)	0.171
Unjudged rate (no annotation)	0.926
Retrieved documents	2400
Annotated documents retrieved that were not cited	0

Table 11: Calibration of the claim-visible judge (B1). One temperature is fitted on the calibration split; no parameter is fitted on the development or held-out samples. The MiniMax family had no calibration-split run, so its row is absent rather than imputed.

Model family	Evaluation split	n	Brier	NLL	ECE	temperature
GPT-6 Luna	calibration	100	0.307	0.586	0.021	0.4
GPT-6 Luna	dev300	300	0.273	0.531	0.047	0.4

reproducible; a sample of the rows was then read manually to check that the automatic classes describe the actual failure. Table 13 gives the distribution over conditions, and Figure 8 separates the development counts by type.

7.9 Case studies

The case records show how the aggregate errors arise. Table 14 in Appendix E records three rows selected deterministically from the saved predictions: one where the two evidence scopes disagreed, one where both scopes were unsure and the validator lost a claim the baseline got right, and one where the baseline endorsed a claim the gold does not support. The second row shows the cost most directly: the two readers agreed that the evidence was inconclusive, the gold annotation says the claim is supported, and no amount of staging recovers it. That is the coverage cost on a single claim.

The dominant class is scope conflict: 38 of V's 87 development errors and 45 of its 100 held-out errors sit on claims where the scopes disagreed, and V abstains on every one by construction. Execution failures are counted separately as an execution status: the development run recovered all of them by retry, and the held-out run, which does not retry, lost 38 of 300 claims to them. We also read a sample of the *successful* rows, because a system can be accidentally right: several V successes are cases where the D_0 citation failed to support the claim while D_1 supplied better evidence.

8 Discussion

Independent retrieval is not sufficient. In both model families on the development sample, and again for the Luna family under the freeze, the staged validator did not significantly reduce false endorsement relative to a baseline that already read the original citations. The 95% interval on the primary comparison includes zero in all three comparisons we ran.

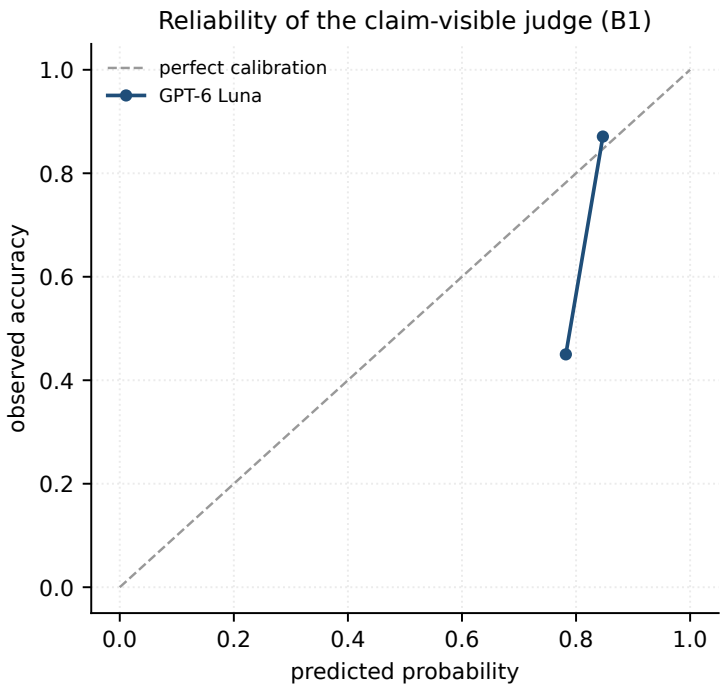


Figure 5: For B1, observed accuracy against verbalised confidence after calibration. Only bins with at least three claims are drawn; this is not a calibration result for the staged validator.

Staging trades recall for precision. The sealed reader makes the system conservative: it endorses less, and what it endorses is right more often. Reported as F1 this is a regression; reported as a precision–coverage curve it is a usable operating point. Both reports are needed, and readers of verification papers should demand both.

The main obstacle is disagreement between scopes, not missing evidence. Most underdetermined verdicts come from two readers of the same corpus reaching different conclusions, not from an empty retrieval pool. Improving retrieval alone will not fix this. A better decision procedure for the conflict case would help, for example evidence chain evaluation [Yang, 2026], scope-matched adjudication, or explicit conflict reporting to the reader.

We can say nothing about human benefit. Explanations exist in the design, and prior work shows that explanations and confidence interact with human calibration [Zhang et al., 2020], but we ran no user study. Appendix D gives the protocol so that the study can be run without redesigning it.

9 Limitations

- **No human annotation.** SciFact's three-way gold cannot score the four-way distinction; the unaddressed/underdetermined split is descriptive. Every number here is a number about label agreement with a crowd-sourced benchmark, not about scientific truth.
- **One corpus, one domain.** Biomedical abstracts, English only, abstract-level text. Claims that require full-text methods, figures, or supplementary material are out of scope, and the corpus bounds every support claim we make.
- **Supplied claims, not generated reports.** The design includes a proposer that turns a report into atomic claims, but the evaluation scores crowd-sourced SciFact claims, not claims proposed from agent-written research reports. A multi-hop agentic track with human reference inventories and adjudicated annotation was planned and not run, so the results speak to checking a claim whose evidence set is given, not to end-to-end report verification.

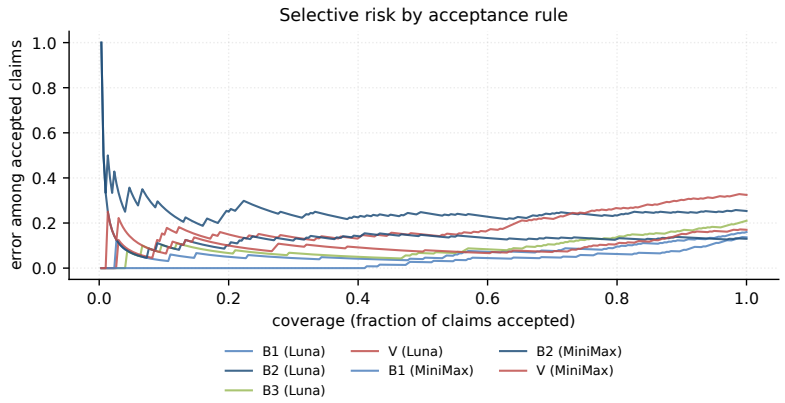


Figure 6: Error among accepted claims as a function of coverage. The acceptance rule differs by method and is stated in the artifact: for V a claim is accepted when the D0 and D1 verdicts agree, for B2 when the sealed reader cited at least one document, and for B1 and B3 when the verbalised confidence is high. The curves therefore compare acceptance policies, not calibrated probabilities.

Table 12: Calls per claim, measured tokens per call, and per-call latency for the Luna family. Calls per claim is fixed by the design. Tokens per call is the run's recorded total divided by its recorded calls, which survives a resumed run; per-claim cost is the product of the first two columns. Each row is taken from the run that recorded the most calls for that condition, which for V is the held-out run: the development sidecar retains only the calls of its final resume pass. The MiniMax family has no cost row; its harness preamble is included in every call and the endpoint that served it was withdrawn before a cost probe could be run.

Method	measured on	calls/claim	tokens/call	tokens/claim
GPT-6 Luna				
B1	dev	1	752	752
B2	dev	2	605	1,209
B3	dev	1	3,072	3,072
V	dev	3	2,579	7,737
O	dev	1	380	380

- **Abstract-level evidence.** Retrieval and reading operate on abstracts. Some claims are not decidable from an abstract, which should appear as unaddressed but may be read as underdetermined.
- **Model families are not matched.** The two families differ in size, training, and harness. The transfer result is about the *direction* of the effect, not its magnitude, and the second family's harness preamble is included in every request, so its token counts are not comparable with the Luna family's.
- **The cross-model sweep is descriptive.** The thirty-claim subset is balanced by gold label and frozen before any model was run, but it is too small to resolve a false-endorsement difference of one percentage point; most paired intervals collapse to zero. It supports a statement about the direction of the shift across models, not about its size, and it does not reproduce the SUPPORT-recall gap measured on the full sample. The rows are the models we could reach without charge or within the spend cap, not a stratified sample of the field, and the closed-weight rows share a training lineage, so they bound sensitivity to model variant rather than to training pipeline.
- **The MiniMax family's matrix is incomplete.** Its B3 condition stopped partway through the sample and its B0, O and calibration conditions never ran, because the free endpoint that served the model was withdrawn and the remaining requests return a model-unavailable error. The MiniMax family therefore corroborates the preregistered V-versus-B2 contrast and the class breakdown, not the full factorial, and we report its missing rows as absent rather than imputing them.
- **One frozen held-out run.** The held-out sample was run once under the freeze record, so its interval is a single-run estimate and the two families were not both run on it.

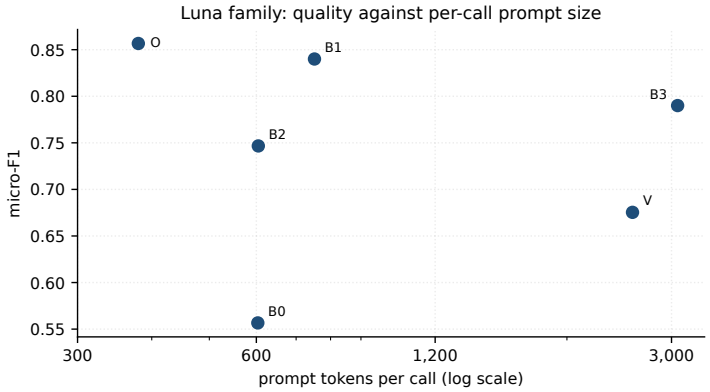


Figure 7: Verification quality against per-call prompt size, Luna family. Cost is measured per call rather than per claim because resumed runs keep only the last invocation's usage record. The MiniMax family is not plotted: its harness preamble dominates every prompt, so a token axis would compare harnesses rather than tasks.

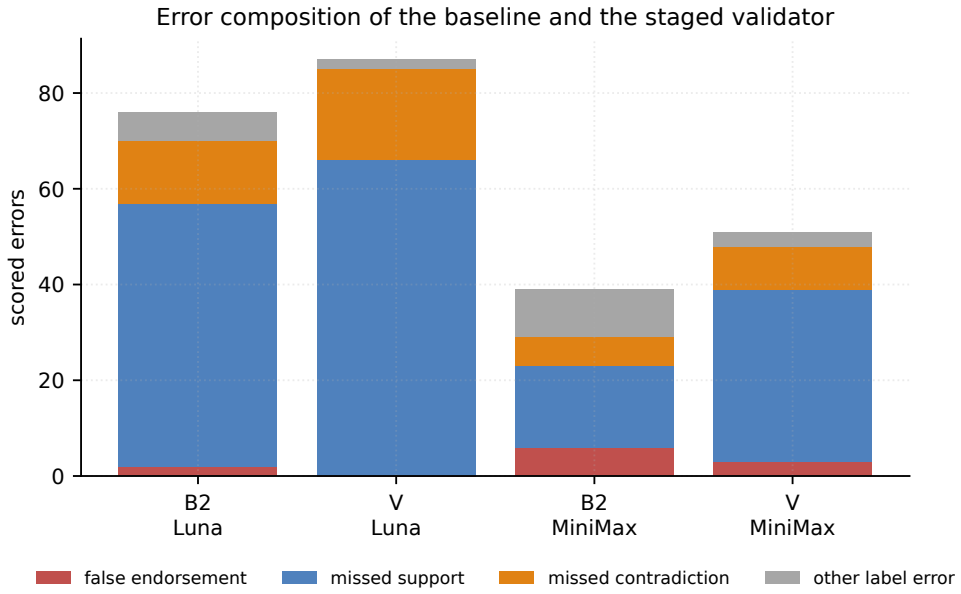


Figure 8: Scored error counts for B2 and V on the development sample, separated by failure type. Execution failures are excluded and counted in the main tables.

- **No matched-budget control was run.** V spends several times the tokens of B2, so compute is a live rival explanation for part of the F1 gap. The planned control repeats B2's comparison step at k samples and votes, matching V's token budget; it is the first experiment we would run next. We report the gap with that caveat rather than claiming it is explained.
- **Retrieval is BM25 only.** Dense retrieval and reciprocal-rank fusion were prepared but not enabled in the product path. A stronger retriever could change the D0/D1 overlap numbers.
- **No user study.** See RQ3 and Appendix D.

10 Ethics and Broader Impact

The task is interpreting published abstracts, not giving medical advice, and the system is evaluated only on a frozen research corpus. The rendered output is a label, a rationale, and a search scope; it is not a clinical recommendation and the interface says so.

Table 13: Error taxonomy over every scored condition. Counts are rows whose predicted label differs from the gold label; execution failures are excluded and reported in the main tables. "held" is the held-out sample and "calibration" the 100-claim calibration split. The MiniMax family has only its development conditions for the reasons given in Section 9.

Method	split	false end.	missed supp.	missed contr.	other	errors
GPT-6 Luna						
B0	dev	82	15	4	32	133
B1	dev	9	19	7	13	48
B2	dev	2	55	13	6	76
B3	dev	10	24	10	19	63
O	dev	3	31	9	0	43
V	dev	0	66	19	2	87
B0	devrerun	82	16	7	31	136
B1	devrerun	8	20	6	13	47
B2	devrerun	2	50	13	11	76
B3	devrerun	9	22	9	20	60
O	devrerun	1	27	11	0	39
V	devrerun	0	67	25	2	94
B2	held	3	52	14	3	72
V	held	0	67	31	2	100
B1	calibration	3	6	2	7	18
MiniMax M3.1 Flash						
B1	dev	17	4	0	20	41
B2	dev	6	17	6	10	39
V	dev	3	36	9	3	51

The failure mode we measured is endorsement without evidence. Our system reduces that risk by abstaining, but abstention has its own risk: a reader who sees an abstention may infer that the claim is false, which would be a new error. We therefore keep the scientific verdict ("underdetermined") separate from the system action ("defer to a reviewer"), and we recommend that any deployment display them separately.

The agent harness used for the MiniMax family and for the open-weight rows of the sweep is an interactive coding agent runtime; a research run inside it inherits that runtime's permissions and preamble. We ran it with a fixed preamble, a fixed corpus, and no tool-enabled tasks. The model itself was withdrawn from its free endpoint during the study, which is why the MiniMax family's matrix is partial; we report the conditions that completed and do not fill the rest by substitution.

11 Conclusion

We built a staged, evidence-first validator for scientific claims: a sealed reader, a claim-visible judge over four labels, a citation auditor over the original references, and a reconciler that reports two evidence scopes separately, with fail-closed execution semantics throughout. Against a strong same-evidence baseline, on a frozen corpus, with locked splits, a preregistered primary comparison, and two model families, the validator did not significantly reduce false endorsement. It became a conservative checker whose errors come mostly from disagreement between two readers of the same literature, and the same direction appears when the pairing is repeated across the wider set of models we could run: contradictions are withheld, endorsements are not gained. We report that as a negative result, with the mechanism, the cost, and the artifact needed to re-run it. The next step is a better decision procedure for scope disagreement.

A Prompts and output contracts

The prompts are versioned files under `prompts/`; the run sidecar records each file's SHA-256 and the bundle hash. The contracts are:

> **Reader (claim-blind).** Input: a neutral question and a list of passages with identifiers. Output: `{"answer": str, "cited_doc_ids": [int]}`. The reader is not shown the claim, the asserted answer, the report, or any gold rationale.
>
> **Judge (claim-visible).** Input: the atomic claim, the sealed reader record, and the passages. Output: `{"label": one of supported, contradicted, unaddressed, underdetermined; "rationale_codes": [str]; "cited_span_ids": [str]; "uncertainty_reasons": [str]}`.
>
> **Citation auditor (D0 only).** Input: the atomic claim and the original cited passages. Output: the judge contract plus `{"citations": [{"span_id": str, "adequacy": adequate | inadequate | unverifiable, "defects": [str]}]}`.
>
> **Compare (B2).** Input: the claim, the neutral question, and the sealed reader record. Output: `{"label": SUPPORT | REFUTE | NEI, "rationale": str}`.

The exact text of every prompt, and the parser that enforces each contract, are in the repository. The parsers reject an out-of-set label, a cited identifier that was not supplied, and a missing mandatory field; each rejection produces a failed execution status rather than a repaired label.

B Metric definitions

Let S be the set of scored claims, $g_i \in \{\text{SUPPORT}, \text{REFUTE}, \text{NEI}\}$ the gold label, and p_i the predicted label. The endorsement class is $E = \text{SUPPORT}$.

$$\text{FE}_{\text{gold}} = \frac{|\{i : p_i = E,\ g_i \neq E\}|}{|\{i : g_i \neq E\}|} \qquad \text{FE}_{\text{pred}} = \frac{|\{i : p_i = E,\ g_i \neq E\}|}{|\{i : p_i = E\}|} \tag{1}$$

Micro-F1 pools the confusion matrix over the three classes; macro-F1 averages the three per-class F1 values. Coverage is $|\{i : i \text{ produced a scored prediction}\}|/|S|$. Latency is the median and 95th percentile of the per-call round-trip time recorded by the client; tokens are the sum of prompt and completion tokens reported by the endpoint. The paired bootstrap resamples cited-document families with replacement (2000 resamples) and reports a percentile interval over the resampled difference; a claim enters the comparison only if every compared method produced an execution-ok, gold-labelled row for it, and dropped claims are counted and reported.

C Calibration details

One temperature is fitted on the calibration split by minimising multiclass negative log-likelihood over the class scores of the one condition that verbalises confidence. No parameter is fitted on development or held-out data. Multiclass Brier score is the mean over claims of $\sum_k (p_k - \mathbb{1}[k = y])^2$; ECE uses ten equal-width bins on the predicted probability of the argmax class. The fitted parameters, the class counts, and the raw-versus-calibrated comparison are in the repository. We report that numerical calibration is validated only for the condition that emits scores; for the staged validator, which emits labels only, we report the disagreement-based selective-use analysis instead and state that its numerical calibration is unvalidated.

D Protocol for the deferred human study

No participant was recruited and no ethics review was obtained, so this paper makes no claim about human benefit. The protocol is recorded here so it can be executed without redesign and so that the absence of results is explicit.

Design. Thirty non-expert adults (a feasible range of 24–36), plus four to six pilot participants, would each see twelve short, expert-reviewed items that retain the scientific qualifiers and span supported, contradicted, and unresolved evidence. Each participant sees each item once. Condition assignment is counterbalanced across participants so that item difficulty is not confounded with interface condition: (A) conclusion plus original citations; (B) A plus the validator verdict and its uncertainty; (C) B plus the evidence-linked explanation. Conditions B and C must use the same system verdict and confidence. Four items per condition per participant, with randomised order and recorded assignment.

Outcomes. An initial 0–100 judgement that the conclusion is adequately supported, the same judgement after the assigned interface, a four-way evidence-status choice, one comprehension question, a use-or-defer decision, and response time. Expert gold support is defined as *supported* versus *not established*; "not established" is not described to participants as factual falsity. The primary exploratory outcome is the post-interface Brier score for evidence support comparing C with B while adjusting for pre-interface belief and item effects; secondary outcomes are appropriate acceptance of supported claims, rejection or deferral of unsupported claims, four-way accuracy, and overreliance when the validator is wrong. Naturally incorrect validator outputs are included in a prespecified error stratum, reported separately from natural error prevalence; any intentionally manipulated advice would require ethics review and debriefing.

Analysis. A mixed-effects model with participant and item random effects if the sample supports it, otherwise participant-level paired summaries with explicit limits on item generalisation. Responses are not treated as independent participants. The study is exploratory unless a prospective power analysis supports stronger claims.

E Case records

The three rows below are the case studies discussed in Section 7.9. They are reproduced from the saved predictions without editing, so the claim text, the labels, and the recorded rationale codes can be checked against the artifact.

Table 14: Case studies from the development sample, selected deterministically from the saved predictions. The two scope labels are the D0 and D1 four-way judgments; the mechanism is the rationale code the run recorded.

Case and claim	Gold	B2	V (D0 / D1)	What the record shows
Scope conflict, abstained *Rapid phosphotransfer rates are correlated with histidine kinase regulator specificity.*	SUPPORT	NEI	NEI (underdetermined / supported)	The independent scope finds support the citation scope does not; the reconciler refuses to choose and abstains.
Baseline false endorsement *Beta-band coherence is enhanced for visible stimuli over invisible stimuli.*	NEI	SUPPORT	NEI (supported / unaddressed)	The baseline endorses from the original citations; the validator abstains, so the endorsement is caught but not replaced.
Coverage loss, scopes agree *ALDH1 expression is associated with poorer prognosis for breast cancer primary tumors.*	SUPPORT	SUPPORT	NEI (underdetermined / underdetermined)	Both scopes call the evidence inconclusive; the validator abstains on a claim the baseline and the gold endorse.

F Reproduction

1. Install the package and its development extras, and verify the pinned corpus hash (`data/pins/scifact/`).
2. Start the model endpoint. For the Luna family, set `OPENAI_API_KEY` and `OPENAI_MODEL`. For the second family, start a local agent server and the loopback bridge in `tools/opencode_bridge.py`, then pass `--base-url` and `--model-id` to the runners.
3. Run the conditions on the development sample (`scripts/run_api_dev300.sh` for the Luna family, `scripts/run_minimax_dev300.sh` for the MiniMax family, which also shows the bridge setup). The MiniMax family's registered conditions are the three that completed; its remaining conditions returned a model-unavailable error, as noted in Section 9.
4. Run the held-out sample once with `--frozen-final`; the runner refuses to start unless the model, prompt, configuration, and corpus hashes match `docs/freeze.json`.
5. Reproduce the cross-model sweep with `scripts/make_sweep_sample.py` (builds the frozen thirty-claim subset from the development manifest and the pinned gold release) and `scripts/run_model_sweep.sh` (runs B2 and V for each registered model, resumably, and stops the closed-weight models at a spend cap).
6. Regenerate every table and figure (`scripts/paper/make_tables.py`, `scripts/paper/make_figures.py`) and rebuild the PDF (`scripts/paper/build_pdf.sh`).

Every number in this paper is read from the generated LaTeX fragments under `paper/tables/`, which are produced from the scored metric JSON files under `docs/paper-assets/tables/`. The prose macros in `paper/tables/macros.tex` are generated by the same script, so a number in the text cannot drift from the artifact.

References

Rami Aly, Zhijiang Guo, Michael Schlichtkrull, James Thorne, Andreas Vlachos, Christos Christodoulopoulos, Oana Cocarascu, and Arpit Mittal. FEVEROUS: Fact extraction and VERification over unstructured and structured information. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.

Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. Chain-of-verification reduces hallucination in large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3563–3578, 2024. doi: 10.18653/v1/2024.findings-acl.212.

Shangbin Feng, Weijia Shi, Yike Wang, Wenxuan Ding, Vidhisha Balachandran, and Yulia Tsvetkov. Don't hallucinate, abstain: Identifying LLM knowledge gaps via multi-LLM collaboration. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14664–14690, 2024. doi: 10.18653/v1/2024.acl-long.786. URL https://aclanthology.org/2024.acl-long.786/.

Luyu Gao, Zhuyun Dai, Panupong Pasupat, Anthony Chen, Arun Tejasvi Chaganty, Yicheng Fan, Vincent Zhao, Ni Lao, Hongrae Lee, Da-Cheng Juan, and Kelvin Guu. RARR: Researching and revising what language models say, using language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16477–16508, 2023. doi: 10.18653/v1/2023.acl-long.910. URL https://aclanthology.org/2023.acl-long.910/.

Max Glockner, Yufang Hou, Preslav Nakov, and Iryna Gurevych. Missci: Reconstructing fallacies in misrepresented science. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4372–4405, 2024. doi: 10.18653/v1/2024.acl-long.240. URL https://aclanthology.org/2024.acl-long.240/.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *PMLR*, pages 1321–1330, 2017.

Eran Hirsch, David Wan, Han Wang, Elias Stengel-Eskin, Mohit Bansal, and Ido Dagan. Who is the agent to blame? localizing faithfulness and citation mistakes in agentic deep research, 2026.

Zhuo Li, Yupeng Zhang, Pengyu Cheng, Jiajun Song, Mengyu Zhou, Hao Li, Shujie Hu, Yu Qin, Erchao Zhao, Xiaoxi Jiang, and Guanjun Jiang. MARCH: Multi-agent reinforced self-check for LLM hallucination, 2026.

Potsawee Manakul, Adian Liusie, and Mark Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9004–9017, 2023. doi: 10.18653/v1/2023.emnlp-main.557. URL https://aclanthology.org/2023.emnlp-main.557/.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878, 2024. doi: 10.18653/v1/2024.acl-long.585. URL https://aclanthology.org/2024.acl-long.585/.

Raúl Ortega and Jose Manuel Gomez-Perez. SciClaims: An end-to-end generative system for biomedical claim analysis. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 141–154, 2025. doi: 10.18653/v1/2025.emnlp-demos.11. URL https://aclanthology.org/2025.emnlp-demos.11/.

Qwen-Applications. MARCH: official implementation, 2026. URL https://github.com/Qwen-Applications/MARCH. GitHub repository Qwen-Applications/MARCH.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North*

American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pages 809–819, 2018. doi: 10.18653/v1/N18-1074. URL https://aclanthology.org/N18-1074/.

Juraj Vladika and Florian Matthes. Comparing knowledge sources for open-domain scientific claim verification. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2103–2114, 2024. doi: 10.18653/v1/2024.eacl-long.128. URL https://aclanthology.org/2024.eacl-long.128/.

David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7534–7550, 2020. doi: 10.18653/v1/2020.emnlp-main.609. URL https://aclanthology.org/2020.emnlp-main.609/.

David Wadden, Kyle Lo, Bailey Kuehl, Arman Cohan, Iz Beltagy, Lucy Lu Wang, and Hannaneh Hajishirzi. SciFact-Open: Towards open-domain scientific claim verification. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 4719–4734, 2022. doi: 10.18653/v1/2022.findings-emnlp.347.

Jerry Wei, Chengrun Yang, Xinying Song, Yifeng Lu, Nathan Hu, Jie Huang, Dustin Tran, Daiyi Peng, Ruibo Liu, Da Huang, Cosmo Du, and Quoc V. Le. Long-form factuality in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Dekun Yang. Calibrated selective fact-checking via evidence chain evaluation, 2026.

Jing Yang, Didier Vega-Oliveros, Taís Seibt, and Anderson Rocha. Explainable fact-checking through question answering. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022. doi: 10.1109/ICASSP43922.2022.9747214.

Boxi Yu, Yuzhong Zhang, Liting Lin, Lionel Briand, and Emir Muñoz. Retromorphic testing with hierarchical verification for hallucination detection in RAG, 2026.

Zhenrui Yue, Huimin Zeng, Lanyu Shang, Yifan Liu, Yang Zhang, and Dong Wang. Retrieval augmented fact verification by synthesizing contrastive arguments. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10331–10343, 2024. doi: 10.18653/v1/2024.acl-long.556.

Yunfeng Zhang, Q. Vera Liao, and Rachel K. E. Bellamy. Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 295–305, 2020. doi: 10.1145/3351095.3372852.

Liwen Zheng, Chaozhuo Li, Xi Zhang, Yu-Ming Shang, Feiran Huang, and Haoran Jia. Evidence retrieval is almost all you need for fact verification. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 9274–9281, 2024. doi: 10.18653/v1/2024.findings-acl.551. URL https://aclanthology.org/2024.findings-acl.551/.